

Exercise 2.1 From a Regular Expression to a minimal DFSA

Consider the regular expression: $(a|b)^*ba(a|b)^*$.

1. Construct the NFA from this expression.

2. Construct the DFSA from the NFA.

3. Construct the minimal DFSA.

Exercise 2.2 Flexing

Using a programming language and lexing tool of your choice (like Flex), write a program that reads a subset of XML and identifies the following token types:

- names,
- tokens, and
- string literals.

Your lexer should correctly recognize the tokens in the following XML file:

```
<!-- Library catalog -->
<library>
  <book id="42">
    <title>The Hitchhiker's Guide to the Galaxy</title>
    <author>Douglas Adams</author>
    <year>1979</year>
    <publisher>Pan Books</publisher>
    <genre>Science Fiction</genre>
    <quote>"Don't Panic!"</quote>
  </book>
</library>
```